University of Minho
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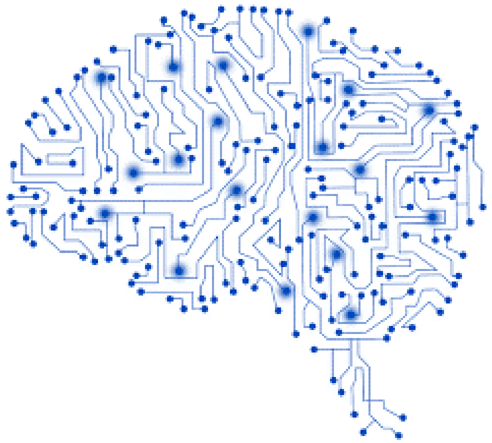

Dados e Aprendizagem Automática

Data Exploration and Preparation

DAA 2025/2026

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Part III

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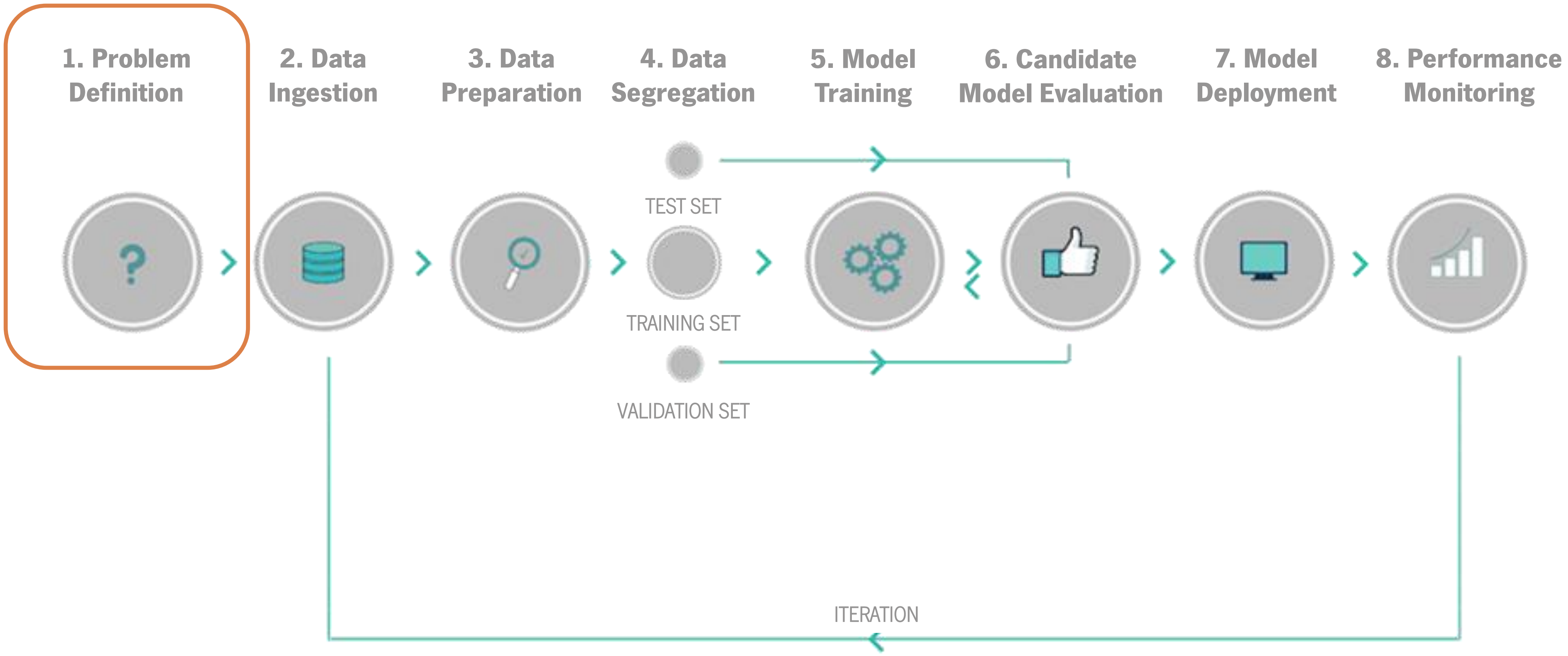
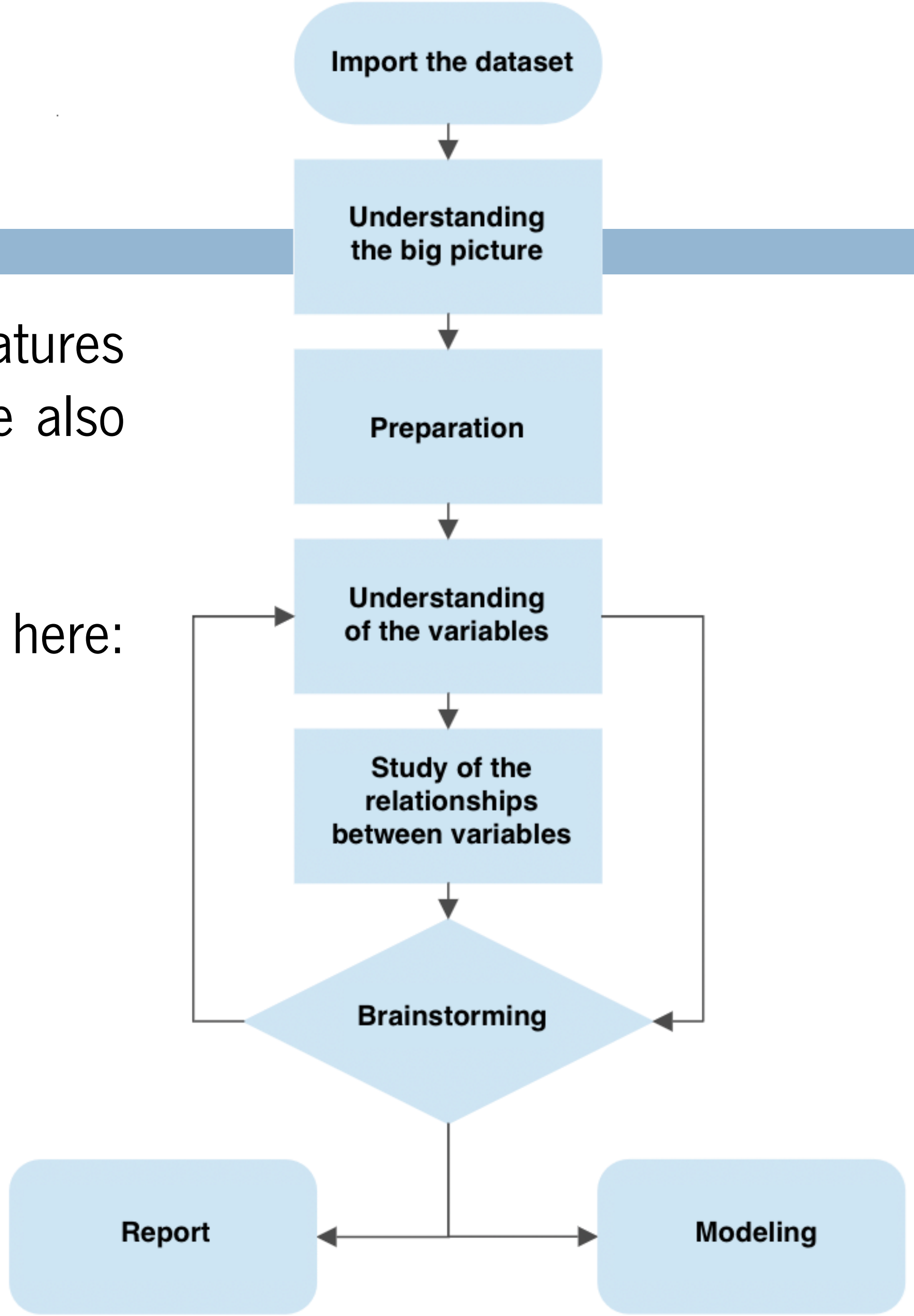


Understanding the Problem

First, we look at the data, to understand the types of features and how important they are to the problem at hand. We also consider whether they are described by another feature.

We are going to use the **wine dataset** available here: https://tinyurl.com/4cshpfac



https://medium.com/@theDrewDag

Imports

Import the necessary libraries:

```
import sklearn as skl
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import preprocessing
```

Load the dataset and inspect some meta-data:

```
df = pd.read_csv('wine.csv')
```

Data Exploration

Data Exploration

What about the actual data? What can we **see**, **learn** or **understand** from it?

```
print(df.columns)
```

```
Index(['Alcohol', 'Malic acid', 'Ash', 'Alcalinity of ash', 'Magnesium',
       'Total phenols', 'Flavanoids', 'Nonflavanoid phenols',
       'Proanthocyanins', 'Color intensity', 'Hue',
       'OD280/OD315 of diluted wines', 'Proline', 'Class'],
      dtype='object')
```

```
df.head()
```

	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	OD280/OD315 of diluted wines	Proline	Class
0	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28	2.29	5.64	1.04	3.92	1065	one
1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050	one
2	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185	one
3	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480	one
4	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735	one

Data Exploration


```
df.tail()
```

	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	OD280/OD315 of diluted wines	Proline	Class
173	13.71	5.65	2.45	20.5	95	1.68	0.61	0.52	1.06	7.7	0.64	1.74	740	three
174	13.40	3.91	2.48	23.0	102	1.80	0.75	0.43	1.41	7.3	0.70	1.56	750	three
175	13.27	4.28	2.26	20.0	120	1.59	0.69	0.43	1.35	10.2	0.59	1.56	835	three
176	13.17	2.59	2.37	20.0	120	1.65	0.68	0.53	1.46	9.3	0.60	1.62	840	three
177	14.13	4.10	2.74	24.5	96	2.05	0.76	0.56	1.35	9.2	0.61	1.60	560	three

```
df.shape
```

```
(178, 14)
```

This dataset contains 178 entries, each with 14 attributes.
The *Class* has **3 classifications**, which refer to the type of wine: one, two and three.

Data Exploration

There are **no null values** and all the attributes are numeric, except for the *Class*.

```
df.dtypes
```

```
Alcohol                         float64
Malic acid                      float64
Ash                             float64
Alcalinity of ash               float64
Magnesium                         int64
Total phenols                   float64
Flavanoids                      float64
Nonflavanoid phenols            float64
Proanthocyanins                 float64
Color intensity                 float64
Hue                             float64
OD280/OD315 of diluted wines    float64
Proline                           int64
Class                            object
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Alcohol                       178 non-null    float64
 1   Malic acid                    178 non-null    float64
 2   Ash                           178 non-null    float64
 3   Alcalinity of ash             178 non-null    float64
 4   Magnesium                     178 non-null    int64
 5   Total phenols                 178 non-null    float64
 6   Flavanoids                    178 non-null    float64
 7   Nonflavanoid phenols          178 non-null    float64
 8   Proanthocyanins               178 non-null    float64
 9   Color intensity               178 non-null    float64
 10  Hue                           178 non-null    float64
 11  OD280/OD315 of diluted wines  178 non-null    float64
 12  Proline                       178 non-null    int64
 13  Class                         178 non-null    object
dtypes: float64(11), int64(2), object(1)
memory usage: 19.6+ KB
```

Data Exploration

We can also obtain descriptive statistics for all numeric data or only the desired data:

```
df.describe()
```

	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	OD280/OD315 of diluted wines	Proline
count	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000
mean	13.000618	2.336348	2.366517	19.494944	99.741573	2.295112	2.029270	0.361854	1.590899	5.058090	0.957449	2.611685	746.893258
std	0.811827	1.117146	0.274344	3.339564	14.282484	0.625851	0.998859	0.124453	0.572359	2.318286	0.228572	0.709990	314.907474
min	11.030000	0.740000	1.360000	10.600000	70.000000	0.980000	0.340000	0.130000	0.410000	1.280000	0.480000	1.270000	278.000000
25%	12.362500	1.602500	2.210000	17.200000	88.000000	1.742500	1.205000	0.270000	1.250000	3.220000	0.782500	1.937500	500.500000
50%	13.050000	1.865000	2.360000	19.500000	98.000000	2.355000	2.135000	0.340000	1.555000	4.690000	0.965000	2.780000	673.500000
75%	13.677500	3.082500	2.557500	21.500000	107.000000	2.800000	2.875000	0.437500	1.950000	6.200000	1.120000	3.170000	985.000000
max	14.830000	5.800000	3.230000	30.000000	162.000000	3.880000	5.080000	0.660000	3.580000	13.000000	1.710000	4.000000	1680.000000

```
df['Color intensity'].describe()
```

```
count    178.000000
mean       5.058090
std        2.318286
min        1.280000
25%        3.220000
50%        4.690000
75%        6.200000
max       13.000000
Name: Color intensity, dtype: float64
```

Data Exploration

What about missing values?

```
df.isna().any()
```

```
Alcohol                         False
Malic acid                      False
Ash                             False
Alcalinity of ash               False
Magnesium                       False
Total phenols                   False
Flavanoids                      False
Nonflavanoid phenols            False
Proanthocyanins                 False
Color intensity                 False
Hue                             False
OD280/OD315 of diluted wines    False
Proline                         False
Class                           False
dtype: bool
```

```
print(df.isna().sum())
```

```
Alcohol                         0
Malic acid                      0
Ash                             0
Alcalinity of ash               0
Magnesium                       0
Total phenols                   0
Flavanoids                      0
Nonflavanoid phenols            0
Proanthocyanins                 0
Color intensity                 0
Hue                             0
OD280/OD315 of diluted wines    0
Proline                         0
Class                           0
dtype: int64
```

Data Exploration

This analysis allows the dataset to be characterised more effectively:

- It contains **178 entries**;
- There are **14 attributes**, 13 of which are physicochemical properties of the wine and one of which is its classification;
- All values are **non-null**;
- There are **no missing values**.

The **goal** when working with this dataset is to **identify the wine type based on its properties**. The target variable categorizes the values as one, two or three.
When used for modelling, these characteristics can be used to **predict the type of wine**.

Data Preparation

Data Preparation

It consists of several steps. Often, you will need to check the API of the library you are using. Here are some useful links for future reference:

- **Numpy** (https://numpy.org/doc/stable/)
- **Pandas** (https://pandas.pydata.org/docs/)
- **Matplotlib** (https://matplotlib.org/stable/users/index.html)
- **Seaborn** (https://seaborn.pydata.org/api.html)
- **Scikit Learn** (https://scikit-learn.org/stable/api/index.html)

Data Preparation

Here are some basic Pandas functions that you may need for data preparation at some point in the future:

```
pandas.DataFrame.drop
pandas.DataFrame.drop_duplicates
pandas.DataFrame.fillna
pandas.DataFrame.isna
pandas.DataFrame.interpolate
pandas.DataFrame.dropna
pandas.DataFrame.groupby
pandas.DataFrame.loc
pandas.DataFrame.iloc
```

Data Preparation

You will also need some basic Sklearn functions and classes:

```
sklearn.preprocessing.MinMaxScaler
sklearn.preprocessing.StandardScaler
sklearn.preprocessing.KBinsDiscretizer
sklearn.preprocessing.LabelEncoder
sklearn.feature_selection
sklearn.metrics
```

Data Preparation and Transformation

Remove duplicate values:

```
print(df.duplicated().sum())
print(df.drop_duplicates(inplace=True))
print(df.info())
```

```
0
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                         Non-Null Count  Dtype  
---  ------                         --------------  -----  
 0   Alcohol                        178 non-null    float64
 1   Malic acid                     178 non-null    float64
 2   Ash                            178 non-null    float64
 3   Alcalinity of ash              178 non-null    float64
 4   Magnesium                      178 non-null    int64  
 5   Total phenols                  178 non-null    float64
 6   Flavanoids                     178 non-null    float64
 7   Nonflavanoid phenols           178 non-null    float64
 8   Proanthocyanins                178 non-null    float64
 9   Color intensity                178 non-null    float64
 10  Hue                            178 non-null    float64
 11  OD280/OD315 of diluted wines   178 non-null    float64
 12  Proline                        178 non-null    int64  
 13  Class                          178 non-null    object 
dtypes: float64(11), int64(2), object(1)
memory usage: 19.6+ KB
None
```

Rename attributes:

```
df.rename(columns={"OD280/OD315 of diluted wines": "Protein Concentration"}, inplace=True)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Alcohol                178 non-null    float64
 1   Malic acid             178 non-null    float64
 2   Ash                    178 non-null    float64
 3   Alcalinity of ash      178 non-null    float64
 4   Magnesium              178 non-null    int64  
 5   Total phenols          178 non-null    float64
 6   Flavanoids             178 non-null    float64
 7   Nonflavanoid phenols   178 non-null    float64
 8   Proanthocyanins        178 non-null    float64
 9   Color intensity        178 non-null    float64
 10  Hue                    178 non-null    float64
 11  Protein Concentration  178 non-null    float64
 12  Proline                178 non-null    int64  
 13  Class                  178 non-null    object 
dtypes: float64(11), int64(2), object(1)
memory usage: 19.6+ KB
```

Data Preparation and Transformation


```
df_clean = df.drop(df.loc[(df['Ash']<2)| ~(df['Alcalinity of ash']>15)].index)
print(df_clean)
```

As all the variables appear to be physicochemical measures, they could all be useful in defining the segmentation of the wine types.

There is **no reason to remove any columns**.

```
     Alcohol  Malic acid   Ash  Alcalinity of ash  Magnesium  Total phenols  \
0      14.23        1.71  2.43               15.6        127           2.80
2      13.16        2.36  2.67               18.6        101           2.80
3      14.37        1.95  2.50               16.8        113           3.85
4      13.24        2.59  2.87               21.0        118           2.80
5      14.20        1.76  2.45               15.2        112           3.27
..       ...         ...   ...                ...        ...            ...
173    13.71        5.65  2.45               20.5         95           1.68
174    13.40        3.91  2.48               23.0        102           1.80
175    13.27        4.28  2.26               20.0        120           1.59
176    13.17        2.59  2.37               20.0        120           1.65
177    14.13        4.10  2.74               24.5         96           2.05

     Flavanoids  Nonflavanoid phenols  Proanthocyanins  Color intensity   Hue  \
0          3.06                  0.28             2.29             5.64  1.04
2          3.24                  0.30             2.81             5.68  1.03
3          3.49                  0.24             2.18             7.80  0.86
4          2.69                  0.39             1.82             4.32  1.04
5          3.39                  0.34             1.97             6.75  1.05
..          ...                   ...              ...              ...   ...
173        0.61                  0.52             1.06             7.70  0.64
174        0.75                  0.43             1.41             7.30  0.70
175        0.69                  0.43             1.35            10.20  0.59
176        0.68                  0.53             1.46             9.30  0.60
177        0.76                  0.56             1.35             9.20  0.61

     Protein Concentration  Proline  Class
0                     3.92     1065    one
2                     3.17     1185    one
3                     3.45     1480    one
4                     2.93      735    one
5                     2.85     1450    one
..                     ...      ...    ...
173                   1.74      740  three
174                   1.56      750  three
175                   1.56      835  three
176                   1.62      840  three
177                   1.60      560  three

[152 rows x 14 columns]
```

Univariate Analysis

Univariate Analysis

Obtain basic information by iterating through each relevant variable:

```
df['Alcohol'].value_counts()
```

```
Alcohol
13.05    6
12.37    6
12.08    5
12.29    4
12.42    3
        ..
13.72    1
13.29    1
13.74    1
13.77    1
14.13    1
Name: count, Length: 126, dtype: int64
```

```
df['Class'].value_counts(normalize=True)
```

```
Class
two      0.398876
one      0.331461
three    0.269663
Name: proportion, dtype: float64
```

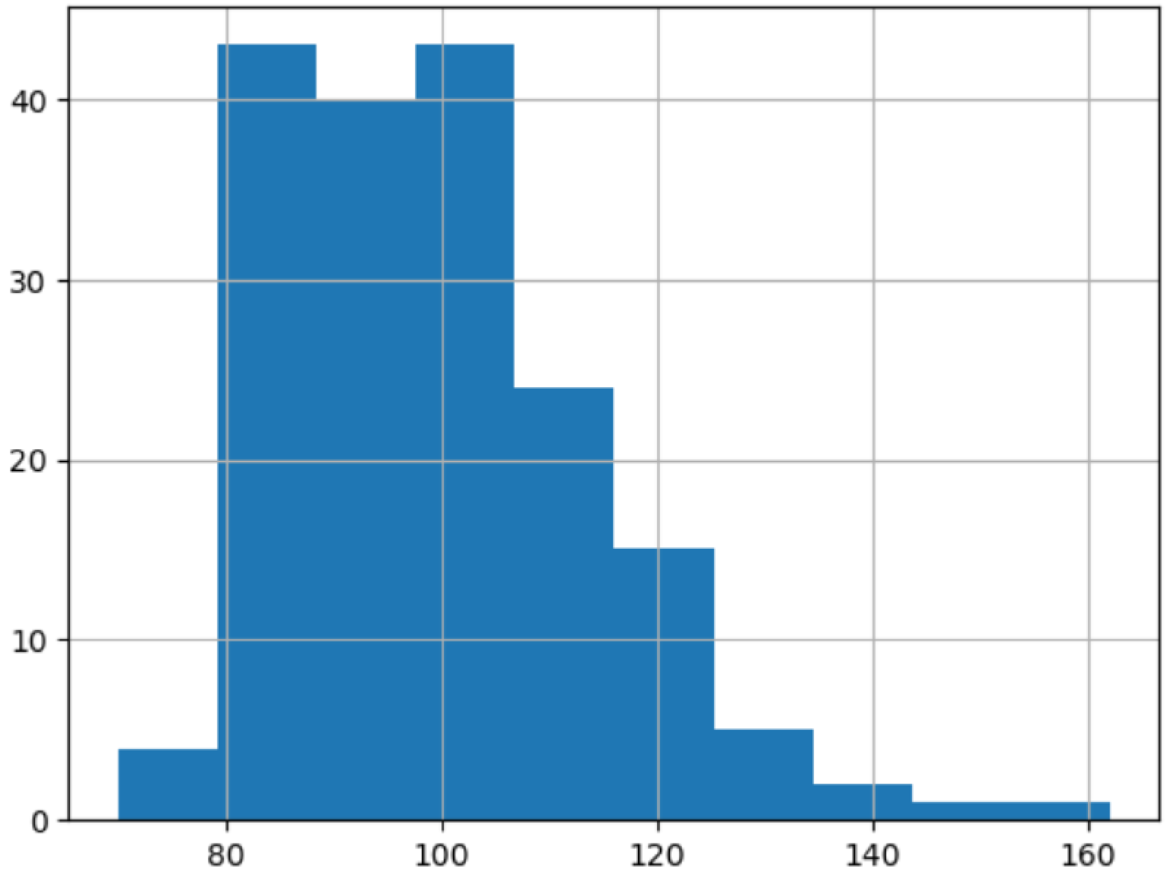
```
'''
Numeric variables
'''
df['Magnesium'].describe()
```

```
count    178.000000
mean      99.741573
std       14.282484
min       70.000000
25%       88.000000
50%       98.000000
75%      107.000000
max      162.000000
Name: Magnesium, dtype: float64
```

Univariate Analysis


```
print(f"Histogram: {df['Magnesium'].hist()}")
```

```
Histogram: Axes(0.125,0.11;0.775x0.77)
```



We observe that it does **not follow a normal curve** and **has spikes**.

Univariate Analysis

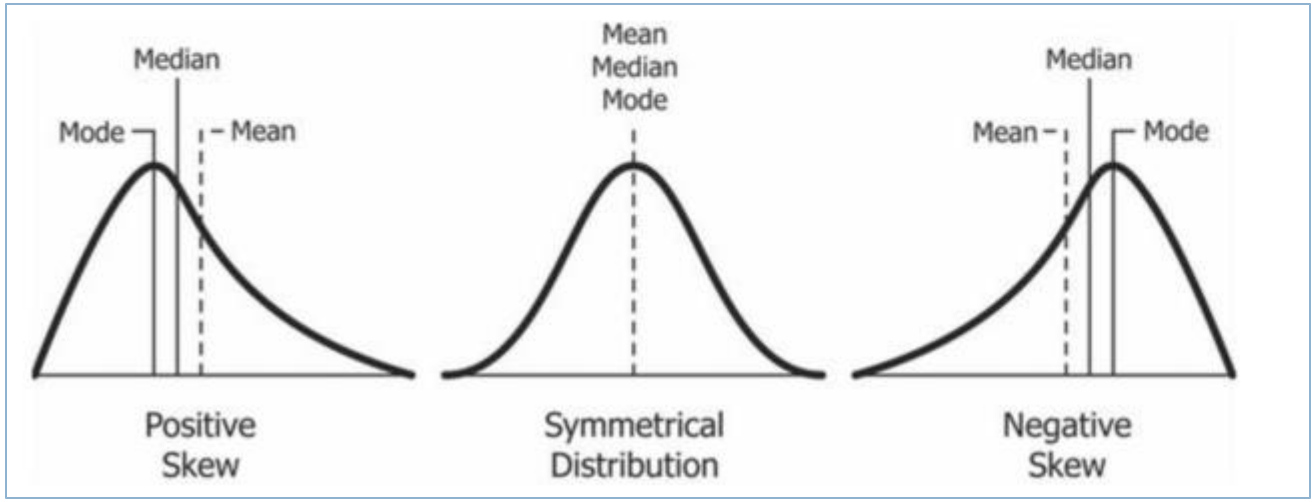

```
print(f"Skewness: {df['Magnesium'].skew()}")
```

```
Skewness: 1.098191054755161
```

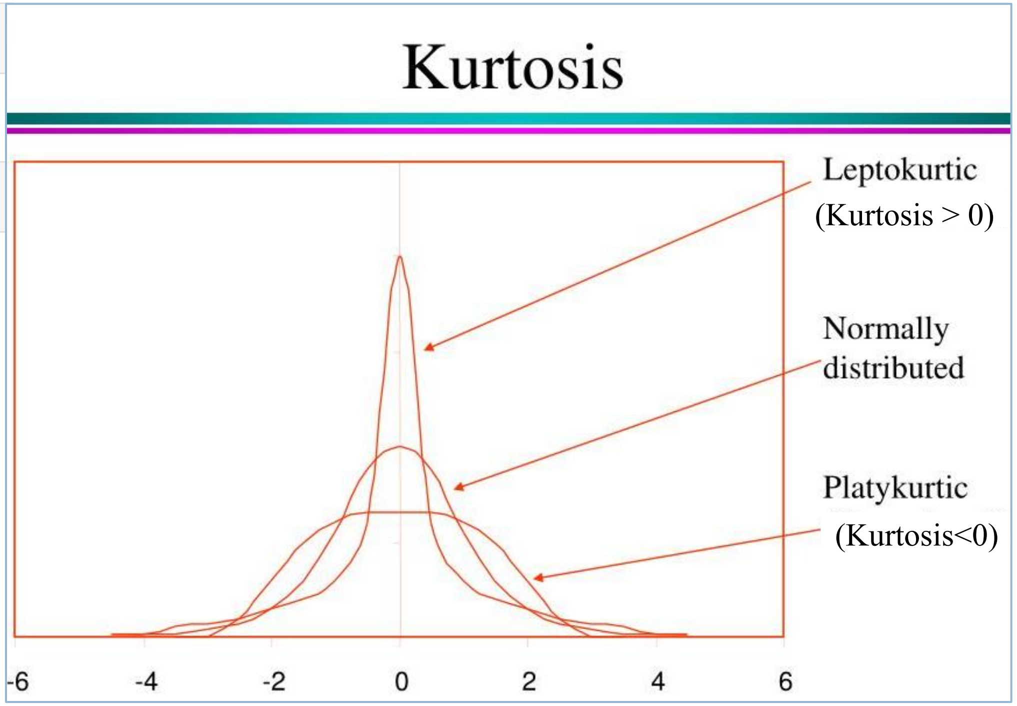
```
print(f"Kurtosis: {df['Magnesium'].kurt()}")
```

```
Kurtosis: 2.1049913235905557
```

The kurtosis and asymmetry values are both **greater than 1**.



https://t.ly/rISUJ



https://t.ly/us-2I

Univariate Analysis

We can now summarize the dataset by creating a short document containing detailed information:

- ***Variable***: the name of the variable;
- ***Type***: the type or format of the variable. This can be categorical, numerical, boolean, etc.;
- ***Context***: useful information to help understand the semantic space of the variable. In the case of our dataset, the context is always physicochemical:
- ***Expectation***: how relevant is this variable for our task? We can use a scale of “High”, “Medium” or “Low”;
- ***Comments***: any comments regarding the variable.

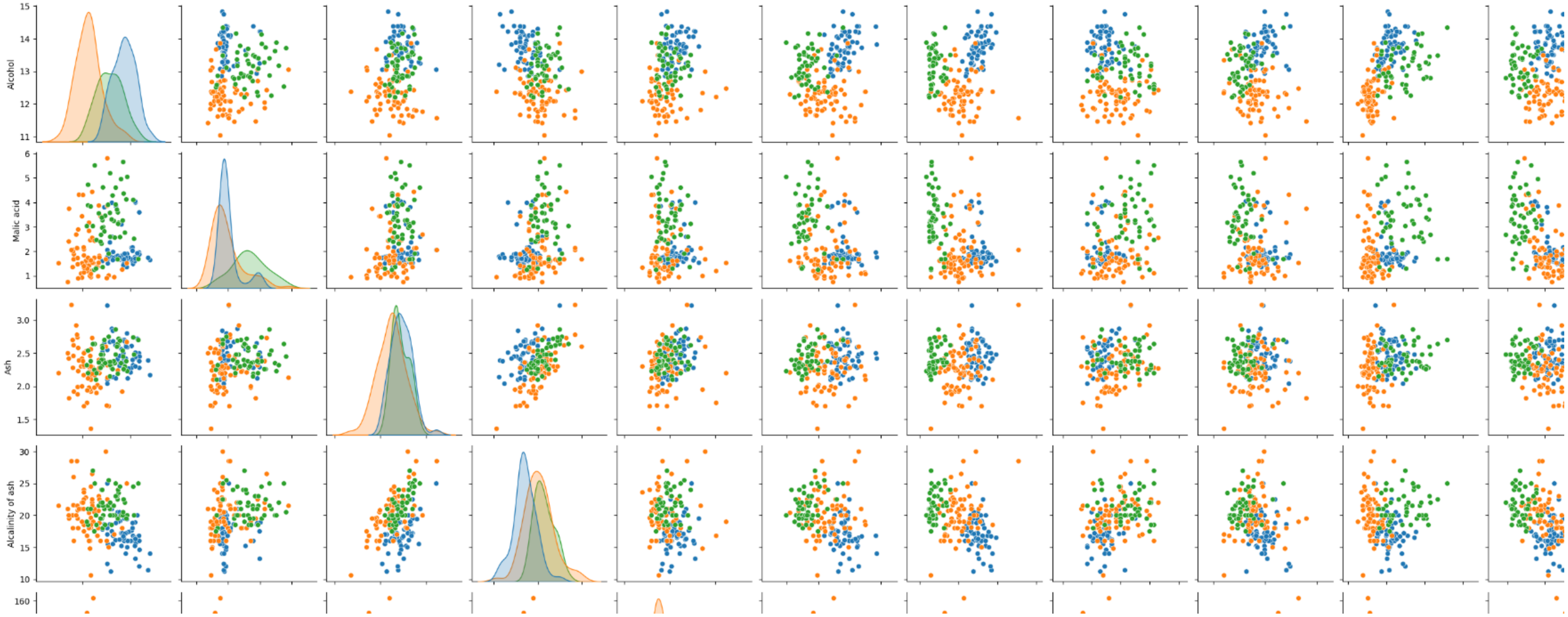
Multivariate Analysis

Multivariate Analysis

We can start by examining at the **relationships** between all the variables:

```
sns.pairplot(df, hue="Class")
```

```
<seaborn.axisgrid.PairGrid at 0x1e31c6eb620>
```



Multivariate Analysis

We can **group** the data by variables:

```
df.groupby(by=['Class']).mean(numeric_only=True)
```

Class	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	Protein Concentration	Proline
one	13.744746	2.010678	2.455593	17.037288	106.338983	2.840169	2.982373	0.290000	1.899322	5.528305	1.062034	3.157797	1115.711864
three	13.153750	3.333750	2.437083	21.416667	99.312500	1.678750	0.781458	0.447500	1.153542	7.396250	0.682708	1.683542	629.895833
two	12.278732	1.932676	2.244789	20.238028	94.549296	2.258873	2.080845	0.363662	1.630282	3.086620	1.056282	2.785352	519.507042

Multivariate Analysis


```
df.groupby(by=['Class', 'Proline']).mean(numeric_only=True)
```

		Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	Protein Concentration
Class	**Proline**												
one	**680**	13.240	3.980	2.290	17.5	103.0	2.640	2.630	0.32	1.66	4.360	0.820	3.00
	735	13.240	2.590	2.870	21.0	118.0	2.800	2.690	0.39	1.82	4.320	1.040	2.93
	760	14.220	3.990	2.510	13.2	128.0	3.000	3.040	0.20	2.08	5.100	0.890	3.53
	770	12.930	3.800	2.650	18.6	102.0	2.410	2.410	0.25	1.98	4.500	1.030	3.52
	780	14.060	1.630	2.280	16.0	126.0	3.000	3.170	0.24	2.10	5.650	1.090	3.71
...	**...**	...	...	...	...	...	...	...	...	...	...	...	...
two	**750**	12.835	0.965	2.155	15.9	123.0	2.215	1.575	0.45	1.59	3.285	1.040	2.12
	870	12.290	1.610	2.210	20.4	103.0	1.100	1.020	0.37	1.46	3.050	0.906	1.82
	886	11.960	1.090	2.300	21.0	101.0	3.380	2.140	0.13	1.65	3.210	0.990	3.13
	937	12.470	1.520	2.200	19.0	162.0	2.500	2.270	0.32	3.28	2.600	1.160	2.63
	985	12.990	1.670	2.600	30.0	139.0	3.300	2.890	0.21	1.96	3.350	1.310	3.50

138 rows × 12 columns

Multivariate Analysis


```
print(df.groupby(by=['Alcohol']).agg([pd.Series.mode]))
```

```
         Malic acid           Ash Alcalinity of ash Magnesium Total phenols  \
               mode          mode              mode      mode          mode   
Alcohol                                                                       
11.03          1.51           2.2              21.5        85          2.46   
11.41          0.74           2.5              21.0        88          2.48   
11.45           2.4          2.42              20.0        96           2.9   
11.46          3.74          1.82              19.5       107          3.18   
11.56          2.05          3.23              28.5       119          3.18   
...             ...           ...               ...       ...           ...   
14.37          1.95           2.5              16.8       113          3.85   
14.38  [1.87, 3.59]  [2.28, 2.38]      [12.0, 16.0]       102   [3.25, 3.3]   
14.39          1.87          2.45              14.6        96           2.5   
14.75          1.73          2.39              11.4        91           3.1   
14.83          1.64          2.17              14.0        97           2.8   

         Flavanoids Nonflavanoid phenols Proanthocyanins Color intensity  \
               mode                 mode            mode            mode   
Alcohol                                                                    
11.03          2.17                 0.52            2.01             1.9   
11.41          2.01                 0.42            1.44            3.08   
11.45          2.79                 0.32            1.83            3.25   
11.46          2.58                 0.24            3.58             2.9   
11.56          5.08                 0.47            1.87             6.0   
...             ...                  ...             ...             ...   
14.37          3.49                 0.24            2.18             7.8   
14.38  [3.17, 3.64]         [0.27, 0.29]    [2.19, 2.96]      [4.9, 7.5]   
14.39          2.52                  0.3            1.98            5.25   
14.75          3.69                 0.43            2.81             5.4   
14.83          2.98                 0.29            1.98             5.2   
```

```
print(df.groupby(by=['Alcohol', 'Flavanoids']).mean(numeric_only=True))
```

```
                    Malic acid   Ash  Alcalinity of ash  Magnesium  \
Alcohol Flavanoids                                                   
11.03   2.17              1.51  2.20               21.5       85.0   
11.41   2.01              0.74  2.50               21.0       88.0   
11.45   2.79              2.40  2.42               20.0       96.0   
11.46   2.58              3.74  1.82               19.5      107.0   
11.56   5.08              2.05  3.23               28.5      119.0   
...                        ...   ...                ...        ...   
14.38   3.17              3.59  2.28               16.0      102.0   
        3.64              1.87  2.38               12.0      102.0   
14.39   2.52              1.87  2.45               14.6       96.0   
14.75   3.69              1.73  2.39               11.4       91.0   
14.83   2.98              1.64  2.17               14.0       97.0   

                    Total phenols  Nonflavanoid phenols  Proanthocyanins  \
Alcohol Flavanoids                                                         
11.03   2.17                 2.46                  0.52             2.01   
11.41   2.01                 2.48                  0.42             1.44   
11.45   2.79                 2.90                  0.32             1.83   
11.46   2.58                 3.18                  0.24             3.58   
11.56   5.08                 3.18                  0.47             1.87   
...                           ...                   ...              ...   
14.38   3.17                 3.25                  0.27             2.19   
        3.64                 3.30                  0.29             2.96   
14.39   2.52                 2.50                  0.30             1.98   
14.75   3.69                 3.10                  0.43             2.81   
14.83   2.98                 2.80                  0.29             1.98   
```

Multivariate Analysis

We can create **bins**.

In **data binning** (or *bucketing*), data is groups into bins (or buckets), whereby values within a small interval are replaced by a single representative value for that interval. This is a **type of data pre-processing** that also includes handling missing values, formatting, normalisation and standardisation.

Binning is a **data smoothing technique**.

Data smoothing is used to **remove noise** from data.

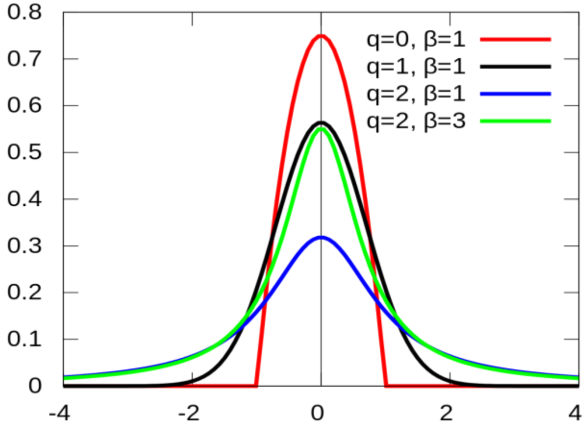
```python
estimator = preprocessing.KBinsDiscretizer(n_bins=3, encode='ordinal', strategy='quantile')
df['alcohol_binned'] = estimator.fit_transform(df[['Alcohol']])

print('Bin Edges')
print(estimator.bin_edges_[0])
print('Alcohol Groups')
print(df.groupby(by=['alcohol_binned']).count())
```

```
Bin Edges
[11.03 12.52 13.48 14.83]
Alcohol Groups
                Alcohol  Malic acid  Ash  Alcalinity of ash  Magnesium  \
alcohol_binned
0.0                  59          59   59                 59         59
1.0                  58          58   58                 58         58
2.0                  61          61   61                 61         61

                Total phenols  Flavanoids  Nonflavanoid phenols  \
alcohol_binned
0.0                        59          59                    59
1.0                        58          58                    58
2.0                        61          61                    61

                Proanthocyanins  Color intensity  Hue  Protein Concentration  \
alcohol_binned
0.0                          59               59   59                     59
1.0                          58               58   58                     58
2.0                          61               61   61                     61

                Proline  Class
alcohol_binned
0.0                  59     59
1.0                  58     58
2.0                  61     61
```

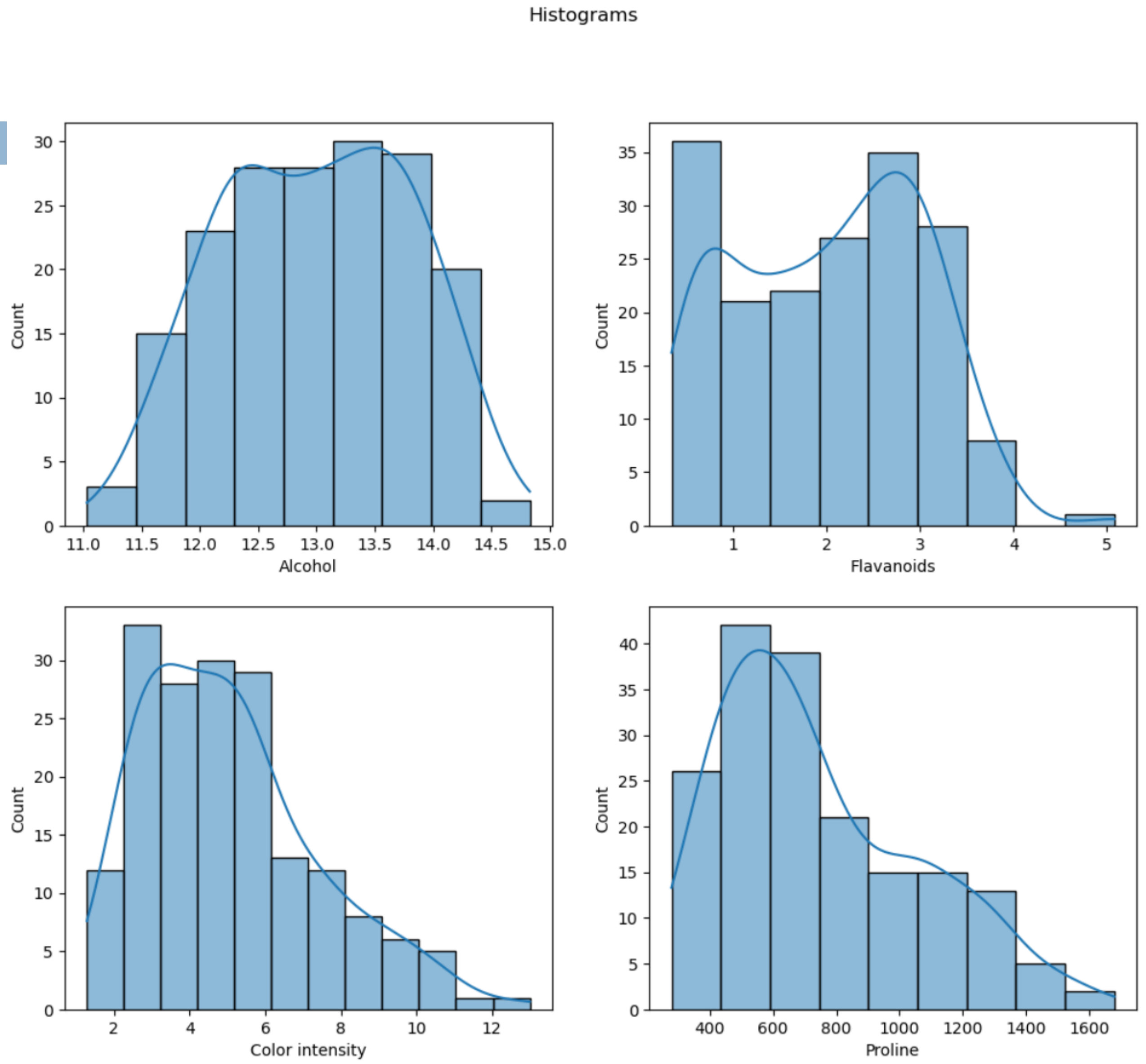
Multivariate Analysis

Dispersion: does it follow a Gaussian distribution?

```
fig, axs = plt.subplots(2, 2, figsize=(12, 10))
fig.suptitle('Histograms')

sns.histplot(df['Alcohol'], ax=axs[0, 0], kde=True)
sns.histplot(df['Flavanoids'], ax=axs[0, 1], kde=True)
sns.histplot(df['Color intensity'], ax=axs[1, 0], kde=True)
sns.histplot(df['Proline'], ax=axs[1, 1], kde=True)
```



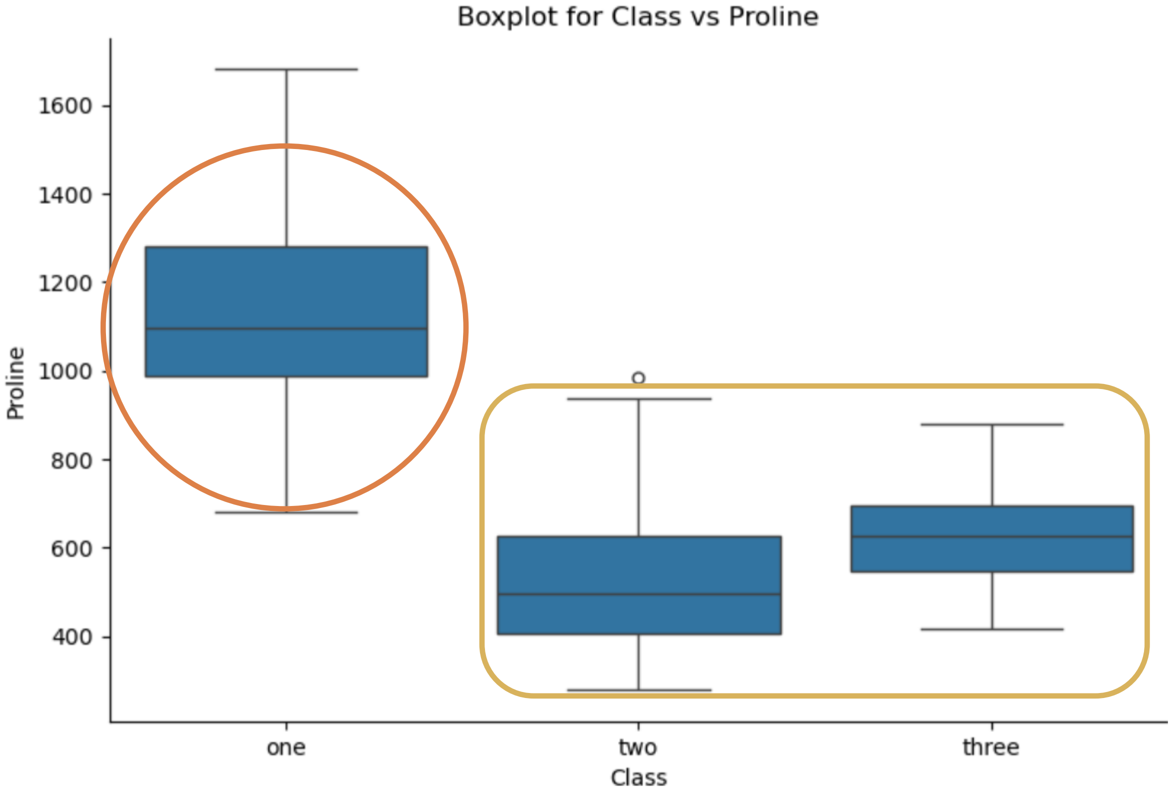
https://en.wikipedia.org/wiki/Q-Gaussian_distribution



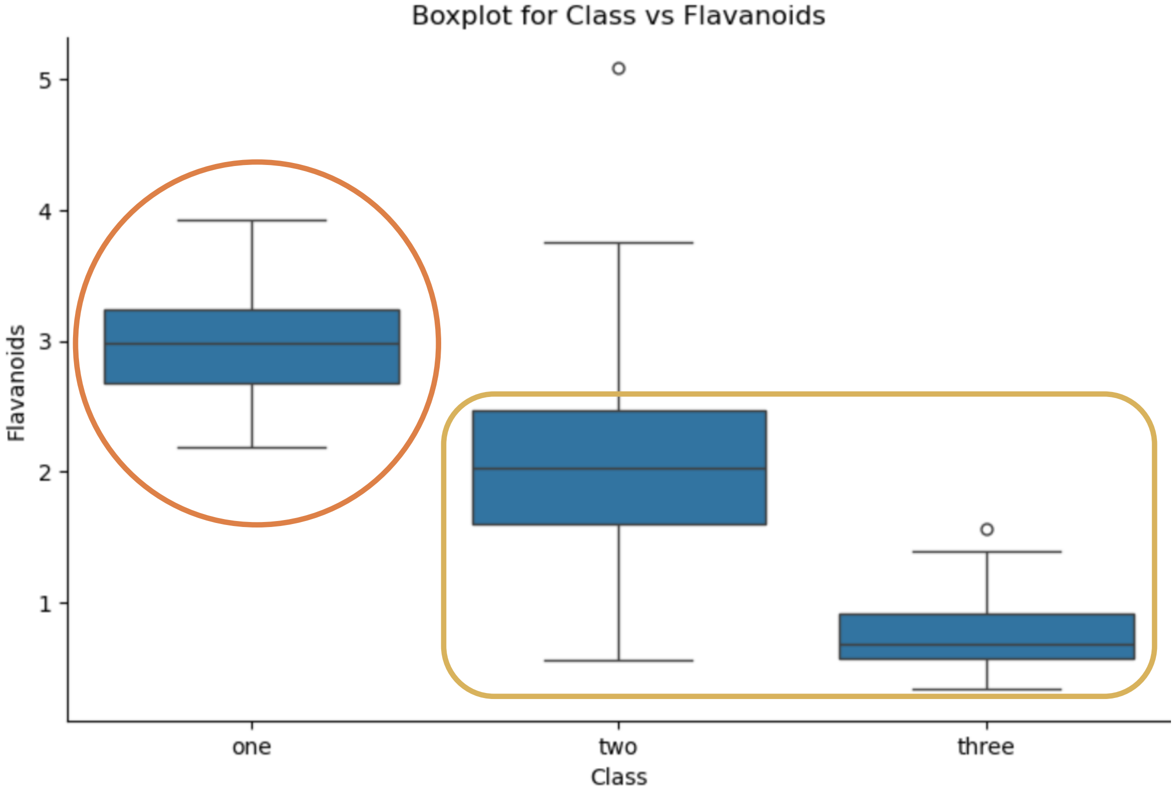
Multivariate Analysis

The best way to understand the relationship between a numeric variable and a categorical variable is through a boxplot:

```
sns.catplot(x="Class", y="Proline", data=df, kind="box", aspect=1.5)
plt.title("Boxplot for Class vs Proline")
plt.show()
```

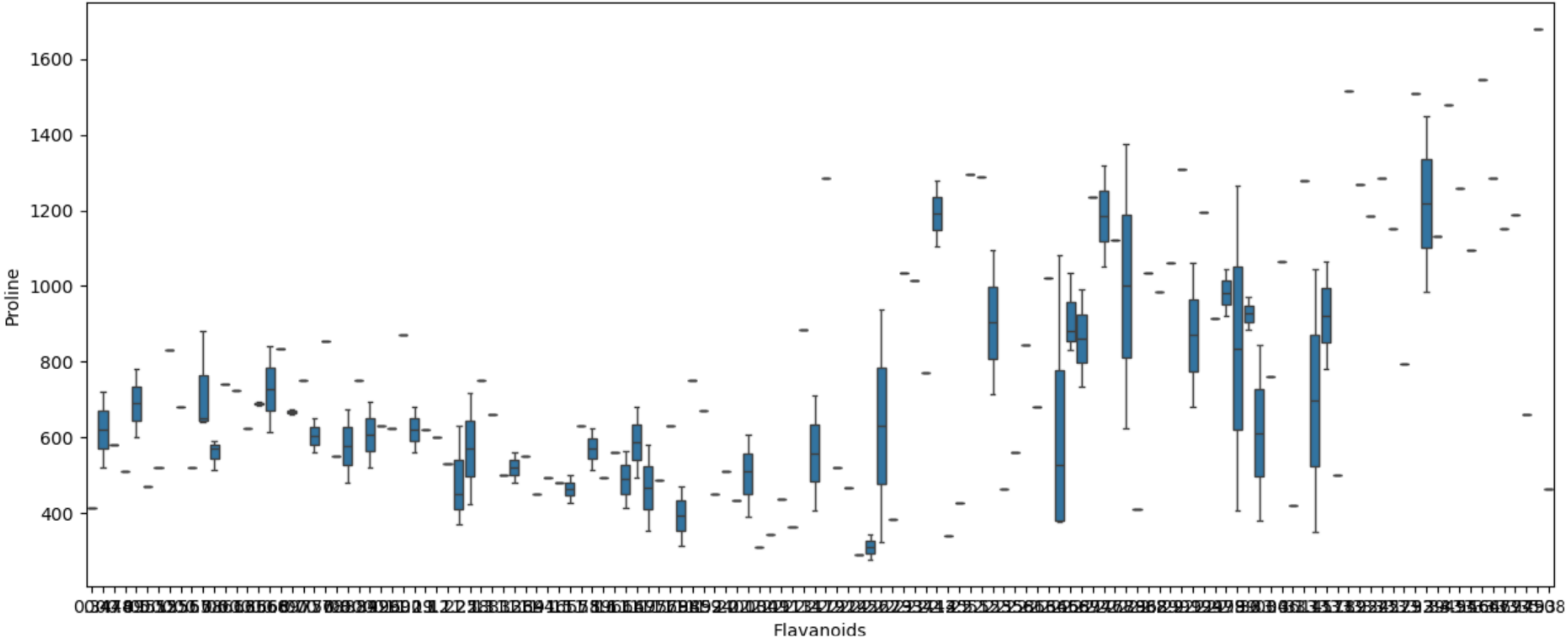



```
sns.catplot(x="Class", y="Flavanoids", data=df, kind="box", aspect=1.5)
plt.title("Boxplot for Class vs Flavanoids")
plt.show()
```



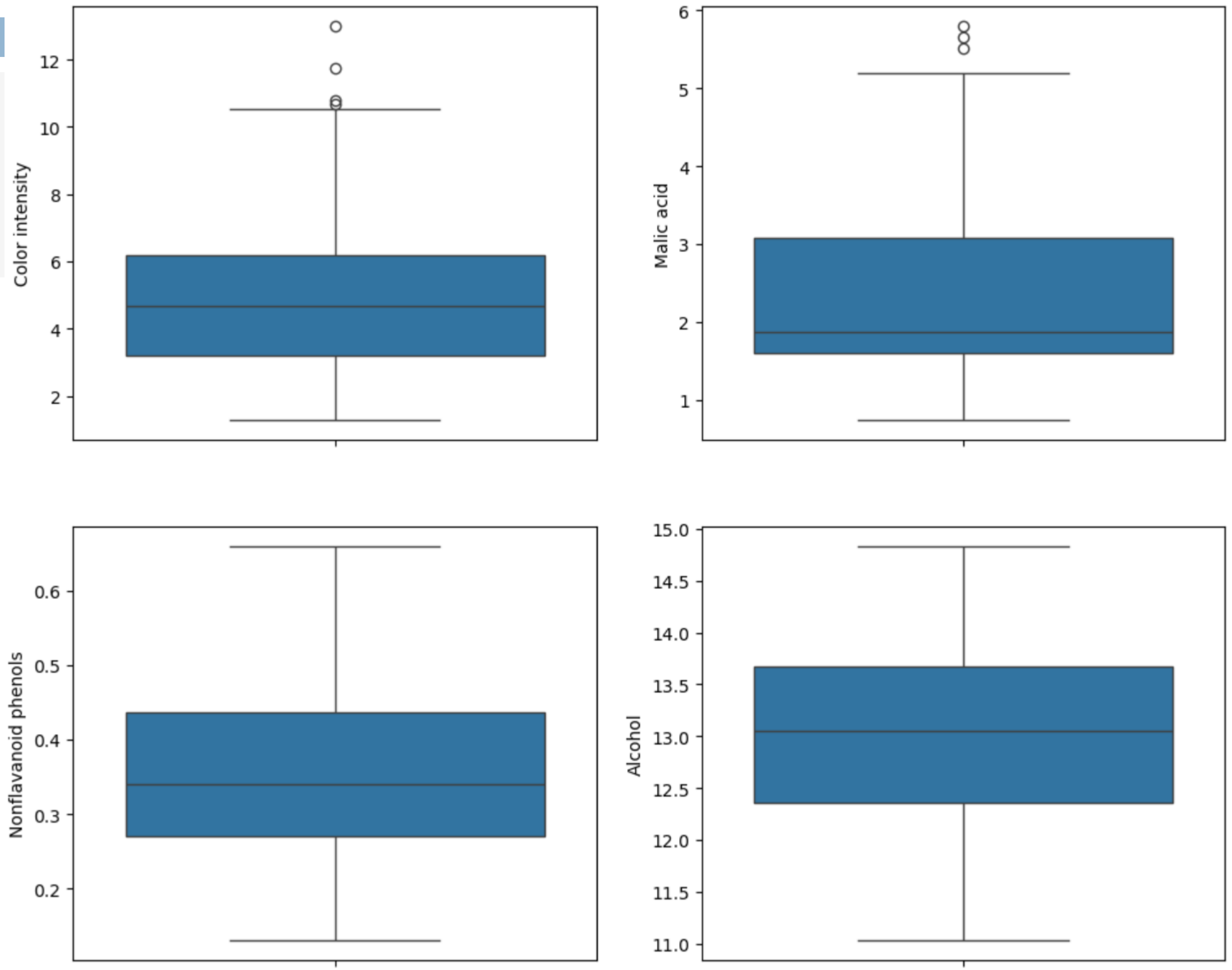
Multivariate Analysis


```
_, ax = plt.subplots(figsize=(15, 6))
fig.suptitle('Boxplot for Flavanoids vs Proline')
sns.boxplot(x=df["Flavanoids"], y=df["Proline"])
```



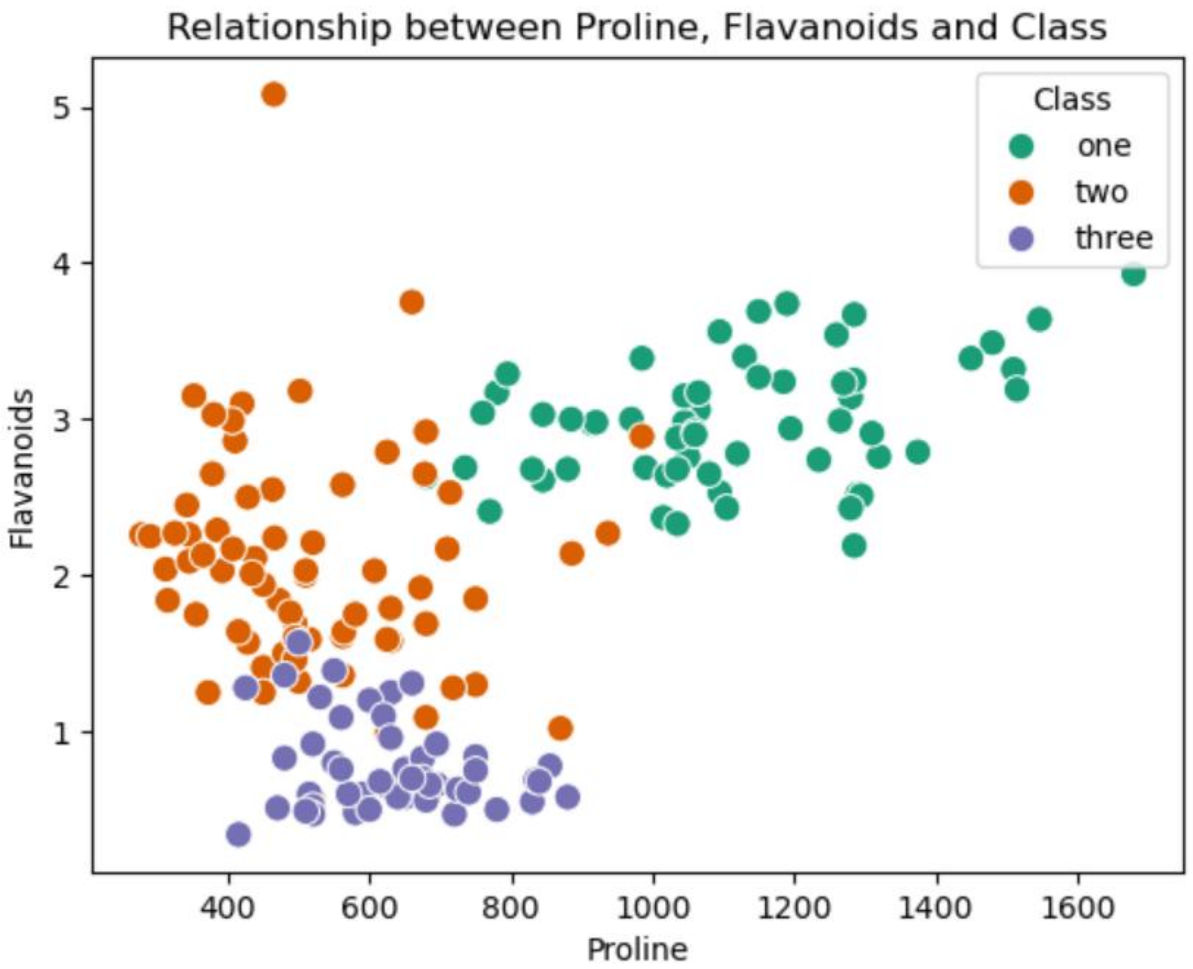
Multivariate Analysis


```
fig, axs = plt.subplots(2, 2, figsize=(12, 10))
fig.suptitle('Boxplots for 4 variables')
sns.boxplot(y=df['Color intensity'], ax=axs[0, 0])
sns.boxplot(y=df['Malic acid'], ax=axs[0, 1])
sns.boxplot(y=df['Nonflavanoid phenols'], ax=axs[1, 0])
sns.boxplot(y=df['Alcohol'], ax=axs[1, 1])
```



Multivariate Analysis

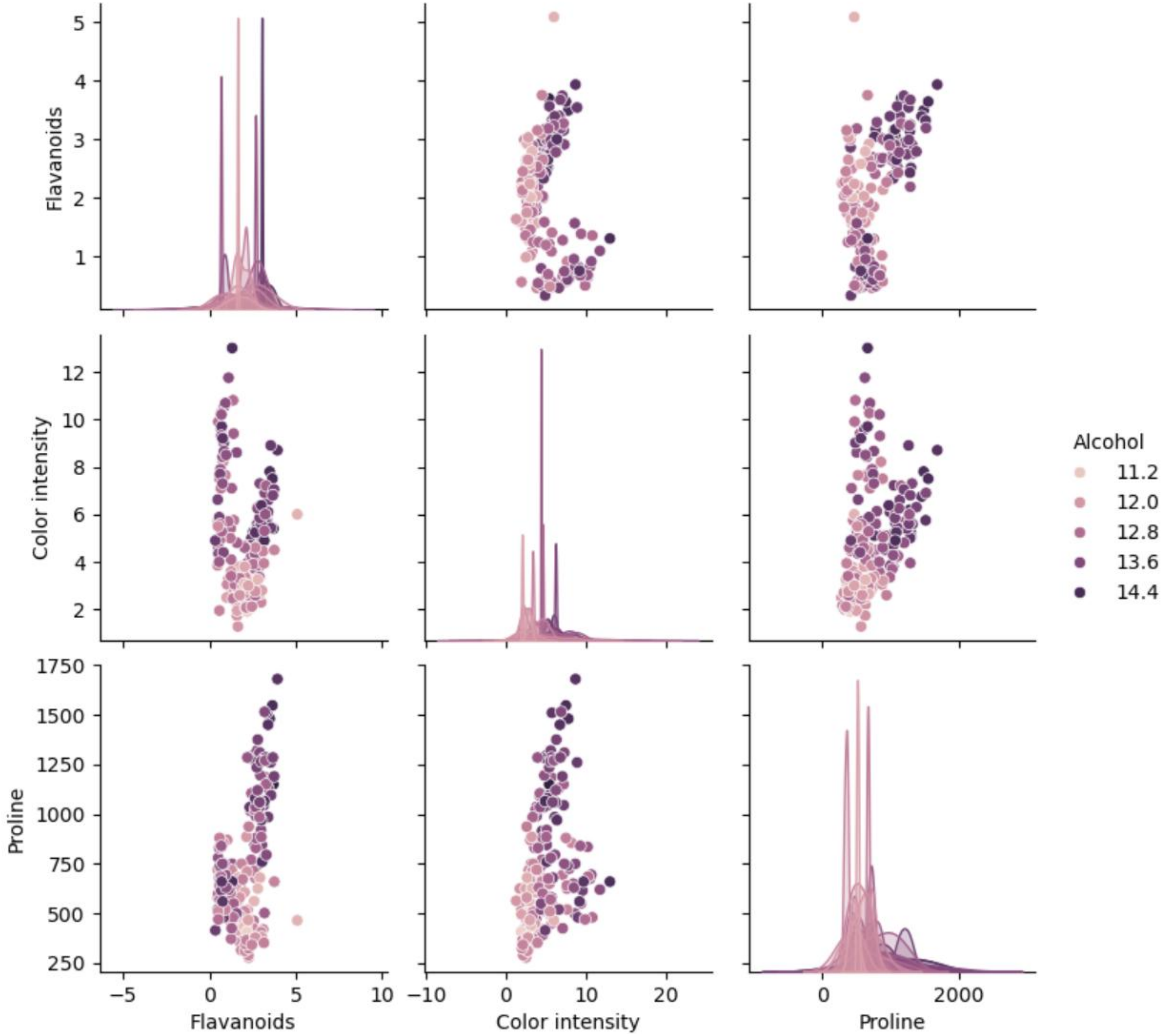

```
sns.scatterplot(x="Proline", y="Flavanoids", hue="Class", data=df, palette="Dark2", s=80)
plt.title("Relationship between Proline, Flavanoids and Class")
plt.show()
```



In *Class* one, *Proline* levels are much **higher**, while *Flavanoid* levels are **stable** at around 3.

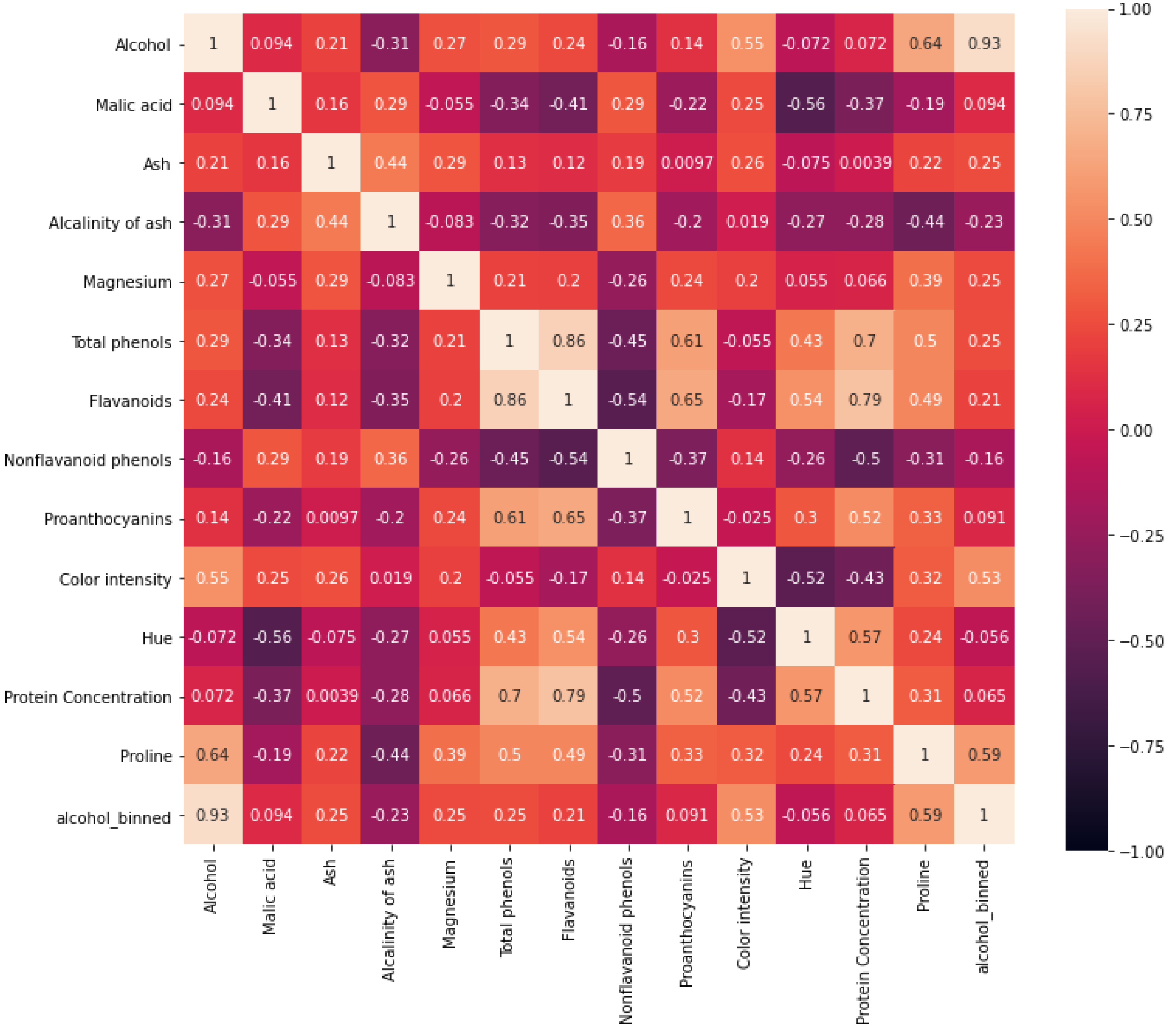
Multivariate Analysis


```
cols = ['Alcohol', 'Flavanoids', 'Color intensity', 'Proline']
_ = sns.pairplot(df[cols], hue='Alcohol', height = 2.5)
```



Multivariate Analysis


```
corr_matrix = df.corr(numeric_only=True)
f, ax = plt.subplots(figsize=(6,4))
sns.heatmap(corr_matrix, vmin=-1, vmax=1, square=True, annot=True)
```



Multivariate Analysis

When the *Class* variable **decreases** (tending towards 0), *Flavanoids*, *Total phenols*, *Proline* and other proteins tend to **increase**.
The opposite is also true.

There is a **very strong correlation between *Alcohol* and *Proline***. High levels of *Alcohol* correspond to high levels of *Proline*.

Critical Analysis

- What are the key characteristics that define the different types of wine?
- Which is the most important component?

Hands On